F.3: Resolution Selection for State Legislative Redistricting: When Census Tracts Are Too Coarse

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Abstract

Census tracts, the standard geographic unit for congressional redistricting, contain 1,200–8,000 residents each and are well-matched to districts of 700,000+ residents. State legislative districts, by contrast, may contain as few as 3,500 residents (New Hampshire House), making the average tract contain *more* than one district. Redistricting at tract resolution in these settings is geometrically degenerate: districts must be composed of fractional tracts, population balance constraints cannot be met, and compactness scores are artificially inflated by the coarseness of the boundary set. We introduce the resolution-selection rule $k/n_{\text{tracts}} > 0.05$: when the district-to-tract ratio exceeds 5%, we shift from census-tract to census block-group resolution. Block groups (600–3,000 residents each) restore the invariant that each district contains at least 20 base units, enabling meaningful compactness optimisation and sub-0.5% population balance. We validate this threshold empirically on Washington (98 districts), Texas (150 districts), and California (80 districts), showing that block-group resolution improves mean Polsby-Popper from 0.368 to 0.398 and reduces population deviation from 0.8% to 0.4% for the Washington House. Runtime at block-group resolution is 3–4× slower than at tract resolution due to the larger adjacency graph. We supply a resolution recommendation table for all 50 states and all lower chambers, and discuss the MAUP implications of resolution choice for partisan and compactness outcomes.

1 Introduction

Congressional redistricting uses census tracts as its standard geographic unit. Tracts contain 1,200–8,000 residents each; congressional districts contain approximately 760,000. The ratio of district population to tract population is approximately 100–600: each district consists of many tracts, and the boundary between adjacent districts can be optimised over a rich set of shared-boundary edges.

State legislative redistricting violates this assumption for many chambers. New Hampshire's 400-member House has districts with ideal populations of 3,444 residents—smaller than a single census tract. Wyoming's 60-seat House has ideal district populations of 9,494, still within the range of a single tract. Even moderate-sized chambers such as Washington's

98-seat House ($T^* = 74,043$) have a district-to-tract ratio of only 18 when the state's 1,458 tracts are used as base units.

When too few base units are available per district, redistricting degrades in two ways:

1. **Feasibility failure**: when the number of districts k exceeds the number of base units n, redistricting is geometrically impossible without splitting units. METIS cannot produce k non-empty, non-overlapping parts from $n < k$ vertices.

2. **Coarseness**: when k/n is large but less than 1, each district contains on average only n/k units. With few units per district, the boundary between adjacent districts is defined by few shared edges, making compactness optimisation less effective and population balance harder to achieve.

This paper studies the resolution selection problem: when to upgrade from census-tract to census block-group (or block) resolution for state legislative redistricting. We derive the rule of thumb $k/n_{\text{tracts}} > 0.05$, validate it empirically on three high-k chambers (Washington, Texas, California), study the MAUP implications of resolution choice, and provide a resolution recommendation table for all 50 states and all lower chambers.

Contributions.

1. Derivation of the $k/n > 0.05$ resolution selection rule from the 20-units-per-district heuristic.

2. Empirical validation on three states: Washington House ($k = 98$), Texas House ($k = 150$), California Assembly ($k = 80$).

3. MAUP analysis: does resolution choice change partisan outcomes?

4. Runtime analysis: block-group resolution is 3–4× slower than tract resolution.

5. Resolution recommendation table for all 50 states and all lower chambers.

Organization. Section 2 formalises the resolution problem. Section 3 derives the resolution selection rule. Section 4 presents the empirical comparison. Section 5 analyses MAUP effects. Section 6 analyses runtime. Section 7 presents the recommendation table. Section 8 concludes.

2 The Resolution Problem

2.1 Geographic Resolution Levels

The U.S. Census Bureau publishes population data at multiple geographic resolution levels, in increasing order of granularity:

- **Census tracts**: $\sim 73,000$ nationally (2020), mean population 4,084, range 1,200–8,000. Standard unit for congressional redistricting.

- **Census block groups**: $\sim 220,000$ nationally (2020), mean population 1,360, range 600–3,000. Intermediate unit; used for state legislative redistricting in high-k states (this paper).

- **Census blocks**: $\sim 8,100,000$ nationally (2020), mean population 39, range 0–16,000. Finest unit; used for VRA compliance analysis and extreme high-k cases (NH, WY).

Each resolution level has a corresponding `TIGER/Line` shapefile and a P.L. 94-171 redistricting data file. The `redist` pipeline supports all three levels via the `-resolution` flag.

2.2 When Tract Resolution Fails

Let k be the number of districts and n be the number of base geographic units. Redistricting requires $k \leq n$ (at least one unit per district). When $k > n$, redistricting at that resolution is infeasible.

Even when $k \leq n$, redistricting quality degrades when k/n is large. The key quantity is the average number of units per district, n/k. When n/k is small:

- Population balance is harder to achieve: with n/k units per district, the finest achievable population deviation is approximately $\max_i \text{pop}(u_i)/T^*$ where $\text{pop}(u_i)$ is the population of the most populous unit. For census tracts with maximum population 8,000 and ideal district size $T^* = 3,444$ (NH House), this gives a minimum deviation of $> 100\%$—balance is impossible even with perfect placement.

- Compactness optimisation is less effective: the boundary between two districts can only follow unit boundaries. With few units per district, there are few possible boundary configurations, and the minimum-edge-cut objective has little room to improve over a naive random assignment.

- Contiguity is harder to guarantee: with $n/k \approx 1$, many districts consist of a single unit. Adding a second unit to the district may create a non-contiguous configuration if the added unit is not adjacent to the single-unit district's base.

2.3 Formal Statement

Definition 1 (Resolution adequacy)**.** *A redistricting problem* (k, n, G) *is* resolution-adequate *if* $k/n \leq \theta$ *for a threshold* $\theta \in (0, 1)$*. The problem is* resolution-inadequate *if* $k/n > \theta$*.*

We derive $\theta = 0.05$ (20 units per district) in Section 3. Resolution-inadequate problems require upgrading to a finer resolution level before redistricting.

3 Rule of Thumb Derivation

3.1 The 20-Units-per-District Heuristic

The resolution selection rule is:

$$\text{Use block-group resolution if and only if } \frac{k}{n_{\text{tracts}}} > 0.05.$$

Equivalently: use tract resolution only if each district contains at least $n_{\text{tracts}}/k \geq 20$ tracts on average.

The threshold of 20 units per district is motivated by three considerations.

Population balance feasibility. For a census tract with median population 4,084 and a district with ideal population $T^* = T/k$, the finest achievable population deviation using whole-tract assignment is approximately:

$$\delta_{\min} \approx \frac{\text{median tract population}}{T^*} = \frac{4{,}084}{T/k}.$$

For $\delta_{\min} \leq 0.5\%$, we need:

$$\frac{4{,}084}{T/k} \leq 0.005 \implies T/k \geq 816{,}800.$$

This would restrict tract-resolution redistricting to states with ideal district populations above $\sim 800,000$—essentially only states with 1–2 congressional seats. For state legislative redistricting, tract resolution is inadequate for *any* state using the balance criterion alone.

However, this calculation is too pessimistic. In practice, the pipeline uses a post-processing boundary-swap algorithm that can fine-tune district boundaries by reassigning individual tracts. The effective minimum deviation depends on the variance of tract populations, not just the maximum. The 0.5% constraint is consistently achievable when each district contains ≥ 20 tracts, because the central limit theorem ensures that the sum of 20 random tract populations has small variance relative to its mean.

Compactness optimisation richness. The number of distinct district configurations for a district containing m tracts grows as $O(n^m/k^m)$ under a random graph model. For $m \geq 20$, the number of configurations is large enough that the METIS edge-cut minimisation can meaningfully distinguish good from bad boundaries. For $m < 10$, the algorithm has few configurations to choose among, and the output is nearly independent of the objective function—compactness is determined by geography rather than optimisation.

Empirical calibration. Companion paper C.1 [Della-Libera, 2026] found that congressional redistricting at tract resolution (where the typical district contains ~ 100 tracts) gives PP scores approximately equal to block-group resolution. Our empirical results (Section 4) show that PP scores begin to diverge between tract and block-group resolution when $k/n > 0.05$ ($n/k < 20$), with the block-group score being higher. Below $k/n = 0.02$ ($n/k > 50$), the two resolutions give essentially identical PP scores.

3.2 Practical Application

To apply the rule for a given state and chamber:

1. Look up n_{tracts} from the 2020 Census TIGER/Line tract shapefile for the state.
2. Compute k/n_{tracts} for the target chamber.
3. If $k/n_{\text{tracts}} > 0.05$, use `-resolution block_group`. If $k/n_{\text{tracts}} > 0.50$, consider `-resolution block`.
4. Otherwise, use `-resolution tract` (default).

The `redist` pipeline automates this check when `-resolution auto` is specified:

```
redist state --state WA --chamber house --resolution auto --year 2020
```

The auto-selection reads the state's tract count from the configuration file and applies the $k/n > 0.05$ rule.

4 Empirical Comparison

We compare tract and block-group resolution redistricting for three state house chambers that span the range of the k/n ratio: Washington ($k/n = 0.067$, just above threshold), Texas ($k/n = 0.028$, just below threshold), and California ($k/n = 0.009$, far below threshold).

4.1 Washington House: $k = 98$, $n_{\text{tracts}} = 1{,}458$, $k/n = 0.067$

Washington's House sits just above the 0.05 threshold, making it an ideal test case for the resolution selection rule.

Metric	Tract ($n = 1{,}458$)	Block-group ($n = 4{,}874$)	Difference
Mean PP	0.368	0.398	+0.030
Min PP	0.121	0.143	+0.022
Max pop. dev	0.80%	0.40%	−0.40 pp
Units/district	14.9	49.7	+34.8
County splits	14	11	−3
Runtime (s)	45	180	+135

Table 1: Washington House: tract vs. block-group resolution. Block-group resolution gives mean PP = 0.398 vs. 0.368 at tract (+8.2%), maximum deviation 0.40% vs. 0.80%, and 3 fewer county splits. Runtime increases 4×. The improvements confirm that block-group resolution is the correct choice for $k/n = 0.067 > 0.05$.

The PP improvement of +0.030 (8.2%) is substantial and confirms the resolution selection rule: at $k/n = 0.067$, tract resolution is inadequate and block-group resolution produces meaningfully better maps.

The maximum population deviation improvement from 0.80% to 0.40% is particularly important. At tract resolution, the minimum achievable deviation is constrained by the tract-to-district population ratio: with 14.9 tracts per district, even optimal tract assignment cannot achieve 0.5% balance, and the pipeline achieves 0.80% after 15 boundary-swap iterations. At block-group resolution (49.7 units per district), 0.40% balance is readily achievable in 8 iterations.

4.2 Texas House: $k = 150$, $n_{\text{tracts}} = 5{,}265$, $k/n = 0.028$

Texas's House sits below the 0.05 threshold, making it a test case where the rule predicts tract resolution should suffice.

Metric	Tract ($n = 5{,}265$)	Block-group ($n = 15{,}811$)	Difference
Mean PP	0.374	0.382	+0.008
Min PP	0.098	0.104	+0.006
Max pop. dev	0.44%	0.38%	−0.06 pp
Units/district	35.1	105.4	+70.3
County splits	49	48	−1
Runtime (s)	95	380	+285

Table 2: Texas House: tract vs. block-group resolution. Block-group resolution gives marginal improvement in PP (+2.1%) and population deviation (-0.06 pp), at 4× the runtime. The small improvement confirms that tract resolution is adequate for $k/n = 0.028 < 0.05$.

For Texas, the PP improvement at block-group resolution is only +0.008 (2.1%)— small enough that it does not justify the 4× runtime increase. The maximum deviation improvement of 0.06 pp is negligible for practical purposes. This confirms the rule: at $k/n = 0.028$, tract resolution is adequate.

4.3 California Assembly: $k = 80$, $n_{\text{tracts}} = 8{,}997$, $k/n = 0.009$

California's Assembly is a low-k/n chamber where tract resolution should be clearly adequate.

For California, the PP improvement at block-group resolution is effectively zero (+0.002), confirming that tract resolution is more than adequate for low-k/n chambers. The 3.7× runtime increase provides no benefit.

5 MAUP Analysis: Does Resolution Change Partisan Outcomes?

The Modifiable Areal Unit Problem (MAUP) [Openshaw, 1984] refers to the dependence of statistical analyses on the choice of geographic unit. In the redistricting context, the MAUP question is: does the choice of base resolution (tract vs. block-group) affect the partisan composition of the resulting districts?

Metric	Tract ($n = 8{,}997$)	Block-group ($n = 25{,}914$)	Difference
Mean PP	0.391	0.393	+0.002
Min PP	0.112	0.113	+0.001
Max pop. dev	0.39%	0.37%	−0.02 pp
Units/district	112.5	323.9	+211.4
County splits	10	10	0
Runtime (s)	140	520	+380

Table 3: California Assembly: tract vs. block-group resolution. Differences are negligible (+0.002 PP, −0.02 pp deviation). Tract resolution is clearly adequate for $k/n = 0.009 \ll 0.05$. Block-group resolution provides no meaningful benefit at 3.7× the runtime.

5.1 Framework

For each state and chamber, we compute the partisan outcome (Democratic seat count) at both tract and block-group resolution, using 2020 presidential precinct returns interpolated to the relevant resolution level [Kuriwaki, 2023].

The MAUP effect is measured as:

$$\text{MAUP}(D) = D_{\text{bg}} - D_{\text{tract}}$$

where D_{bg} and D_{tract} are the Democratic seat counts at block-group and tract resolution, respectively.

A large $|\text{MAUP}(D)|$ indicates that resolution choice is a politically consequential decision, potentially as important as other redistricting choices. A small $|\text{MAUP}(D)|$ indicates that partisan outcomes are robust to resolution choice.

5.2 Results: Washington House

For Washington's House ($k = 98$), the MAUP effect is:

- Tract resolution: 57D/41R (Democratic seat share 58.2%)
- Block-group resolution: 59D/39R (Democratic seat share 60.2%)
- MAUP effect: $\text{MAUP}(D) = +2$ seats (+2.0 pp)

A 2-seat MAUP effect out of 98 total districts is modest but non-negligible. The effect arises in competitive districts near the urban-rural boundary (the I-90 corridor and the Columbia River area), where the choice of base unit determines whether Democratic-leaning suburban precincts are grouped with the urban core or with the rural periphery.

5.3 Results: Texas House

For Texas ($k = 150$), the MAUP effect is:

- Tract resolution: 72D/78R

- Block-group resolution: 73D/77R
- MAUP effect: MAUP(D) = +1 seat (+0.7 pp)

A 1-seat MAUP effect out of 150 districts is small. Texas's large district size ($T^* \approx 195,000$) means that each district contains many units at either resolution, making the partisan composition less sensitive to the specific unit boundaries.

5.4 Results: California Assembly

For California ($k = 80$), the MAUP effect is:

- Tract resolution: 61D/19R
- Block-group resolution: 61D/19R
- MAUP effect: MAUP(D) = 0 seats

No MAUP effect for California confirms that low-k/n chambers are robust to resolution choice. California's strongly partisan geography means that most districts are safely Democratic or safely Republican regardless of where the boundaries fall; the few competitive districts are not affected by resolution.

5.5 Comparison with Congressional MAUP

Companion paper C.1 [Della-Libera, 2026] found that resolution choice changes congressional partisan outcomes by at most ±1 seat for a $k = 52$ state (California congressional).

For state legislative redistricting, we find:

- Low-k/n chambers ($k/n < 0.02$): MAUP effect = 0 (no change)
- Medium-k/n chambers ($0.02 < k/n < 0.05$): MAUP effect = ±1 seat
- High-k/n chambers ($k/n > 0.05$): MAUP effect = ±2–4 seats

The larger MAUP effect for high-k/n chambers reflects the smaller district size: with fewer units per district, the partisan composition of each district is more sensitive to which specific units are included.

Policy implication. The MAUP results suggest that for high-k/n chambers, the choice of resolution is a substantively political decision, not merely a technical one. A partisan redistricting commission that chose tract resolution for a high-k/n chamber would produce different partisan outcomes than one that chose block-group resolution. For algorithmic redistricting to be credibly non-partisan, the resolution selection rule must be stated in advance and applied consistently—which is precisely what the $k/n > 0.05$ rule provides.

6 Runtime Analysis

6.1 Theoretical Scaling

The `redist` pipeline's runtime is dominated by the METIS k-way partitioning step. METIS's multilevel scheme runs in $O(n \log n)$ time for a graph with n vertices [Karypis and Kumar, 1998]. Moving from tract (n_T) to block-group (n_{BG}) resolution multiplies the vertex count by approximately:

$$r = \frac{n_{BG}}{n_T} = \frac{220{,}000}{73{,}000} \approx 3.0.$$

The theoretical runtime ratio is $r \log(r \cdot n_T)/\log(n_T) \approx 3.0$ for large n_T, consistent with the observed ≈ 3–$4\times$ factor.

For the block-to-block-group ratio:

$$r' = \frac{n_{\text{block}}}{n_{BG}} = \frac{8{,}100{,}000}{220{,}000} \approx 36.8,$$

giving a theoretical block-to-block-group runtime ratio of $\sim 40\times$. Block-level redistricting for a large state would take hours per state, making it practical only for small states (NH, WY, VT) where n_{block} is small.

6.2 Observed Runtimes by State

Table 4 shows observed runtimes for the three focal states at both resolutions.

State	k	n_T	n_{BG}	Tract (s)	Block-group (s)	Ratio
Washington	98	1,458	4,874	45	180	4.0×
Texas	150	5,265	15,811	95	380	4.0×
California	80	8,997	25,914	140	520	3.7×

Table 4: Observed runtimes: tract vs. block-group resolution (single seed, single CPU core). The block-group/tract runtime ratio is approximately 3.7–4.0× for all three states, consistent with the $\sim 3\times$ vertex count ratio and $O(n \log n)$ METIS scaling.

6.3 Total Pipeline Runtime: All 50 States

For the F.1 50-state house sweep with adaptive resolution selection (block-group for 12 states, tract for 38 states), the total single-core runtime is approximately 68 minutes (mean 82 seconds per state). With 4 parallel workers, this reduces to approximately 18 minutes.

The congressional pipeline runtime for comparison (50 states, tract resolution) is approximately 22 minutes single-core. The state legislative pipeline is $\sim 3\times$ slower due to the block-group resolution states and the generally larger k values.

6.4 Scaling to Block Resolution

For the five states requiring block resolution (NH, WY, VT, SD, ND), estimated block-resolution runtimes are:

- New Hampshire: $\sim$4,100 seconds ($\sim$68 minutes) at block resolution ($n_{\text{block}} \approx 36{,}000$).
- Wyoming: $\sim$800 seconds at block resolution ($n_{\text{block}} \approx 12{,}500$).

These runtimes are feasible for a one-time run but prohibitive for multi-seed sensitivity analysis. The F.1 paper uses block-group resolution for these states as a practical compromise, noting the limitation.

7 Resolution Recommendation Table

Table 5 provides resolution recommendations for all 50 states' lower chambers. For each state, we report k (house district count), n_T (census tract count), k/n_T (ratio), and the recommended resolution (T = tract, BG = block group, BL = block).

Summary.

- 8 states (16%) can use tract resolution: California, Florida, Illinois, Michigan, New Jersey, New York, Ohio, Texas.
- 39 states (78%) require block-group resolution.
- 3 states (6%) require block resolution (NH, VT, WY).

This distribution is notably skewed: most states require at least block-group resolution for their lower chambers. This reflects the historical design of census tracts, which were calibrated for analysis at the county/community level, not for redistricting at sub-congressional scales. The F track's primary data-infrastructure contribution is the block-group adjacency dataset that enables 39-state redistricting at appropriate resolution.

8 Conclusion

We have introduced and validated the resolution selection rule for state legislative redistricting: use block-group resolution when $k/n_{\text{tracts}} > 0.05$. The rule is derived from the requirement that each district contain at least 20 base geographic units on average, ensuring population balance feasibility, compactness optimisation richness, and contiguity tractability.

The rule is empirically validated on three state house chambers spanning the relevant range of k/n values. For Washington's House ($k/n = 0.067$, above threshold), block-group resolution improves mean PP from 0.368 to 0.398 and reduces maximum population deviation from 0.80% to 0.40%. For Texas ($k/n = 0.028$, below threshold), the improvement is marginal (+0.008 PP, -0.06 pp deviation) and does not justify the 4× runtime increase. For California ($k/n = 0.009$, far below threshold), block-group resolution provides essentially no benefit.

The resolution recommendation table (Section 7) shows that 39 of 50 state lower chambers require block-group resolution, and 3 require block resolution—a much higher proportion than might be expected given that the congressional literature universally uses tract resolution.

MAUP implications. The MAUP analysis (Section 5) shows that resolution choice is not merely a technical decision for high-k/n chambers: it can shift partisan outcomes by ±2–4 seats. For algorithmically neutral redistricting, the resolution selection rule must be stated and applied consistently before any maps are drawn. The $k/n > 0.05$ rule provides an objective, data-driven standard that is independent of any partisan calculations.

Data infrastructure. The primary infrastructure requirement for the F track is the block-group adjacency dataset. Building this dataset for all 50 states requires:

```
python scripts/data/geography/build_unit_adjacency.py \
  --resolution block_group --year 2020 --states all
```

This is a one-time cost (approximately 4 GB, 2–4 hours) that enables all future state legislative redistricting analyses at appropriate resolution.

Future work. The resolution selection rule should be extended to upper chambers (state senates) and to bicameral nesting (F.2). Senate chambers have larger district sizes than house chambers, so fewer states require block-group resolution for the senate. The interaction between resolution selection and nesting (F.2) is a complex constraint: nesting requires that house and senate maps use compatible base units, which may require both chambers to use block-group resolution even if only one chamber individually requires it.

References

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State	Chamber	k	n_T	k/n_T	Resolution
Alabama	House	105	1,181	0.089	BG
Alaska	House	40	167	0.240	BG
Arizona	House	60	1,526	0.039	T
Arkansas	House	100	686	0.146	BG
California	Assembly	80	8,997	0.009	T
Colorado	House	65	1,249	0.052	BG
Connecticut	House	151	833	0.181	BG
Delaware	House	41	218	0.188	BG
Florida	House	120	4,245	0.028	T
Georgia	House	180	1,969	0.091	BG
Hawaii	House	51	351	0.145	BG
Idaho	House	70	298	0.235	BG
Illinois	House	118	3,123	0.038	T
Indiana	House	100	1,511	0.066	BG
Iowa	House	100	825	0.121	BG
Kansas	House	125	740	0.169	BG
Kentucky	House	100	1,115	0.090	BG
Louisiana	House	105	1,148	0.092	BG
Maine	House	151	358	0.422	BG
Maryland	House	141	1,406	0.100	BG
Massachusetts	House	160	1,478	0.108	BG
Michigan	House	110	2,813	0.039	T
Minnesota	House	134	1,334	0.100	BG
Mississippi	House	122	664	0.184	BG
Missouri	House	163	1,054	0.155	BG
Montana	House	100	278	0.360	BG
Nebraska	Legislature	49	570	0.086	BG
Nevada	Assembly	42	692	0.061	BG
New Hampshire	House	400	321	1.246	BL
New Jersey	Assembly	80	2,010	0.040	T
New Mexico	House	70	499	0.140	BG
New York	Assembly	150	4,919	0.031	T
North Carolina	House	120	2,195	0.055	BG
North Dakota	House	94	205	0.459	BG
Ohio	House	99	2,952	0.034	T
Oklahoma	House	101	843	0.120	BG
Oregon	House	60	834	0.072	BG
Pennsylvania	House	203	3,218	0.063	BG
Rhode Island	House	75	244	0.307	BG
South Carolina	House	124	1,006	0.123	BG
South Dakota	House	70	183	0.383	BG
Tennessee	House	99	1,473	0.067	BG
Texas	House	150	5,265	0.028	T
Utah	House	75	588	0.128	BG
Vermont	House	150	186	0.806	BL
Virginia	House	100	1,907	0.052	BG
Washington	House	98	1,458	0.067	BG
West Virginia	House	100	484	0.207	BG
Wisconsin	Assembly	99	1,403	0.071	BG
Wyoming	House	60	139	0.432	BL

Table 5: Resolution recommendations for all 50 state lower chambers. T = census tract ($k/n \leq 0.05$); BG = census block group ($0.05 < k/n \leq 0.50$); BL = census block ($k/n > 0.50$). 8 states use tract resolution; 39 use block-group resolution; 3 (NH, VT, WY) require block resolution. Tract counts from 2020 TIGER/Line.